

Adverse Selection I

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Readings: MSPC Chapter 4.1-4.2

Agenda

Overview

- Market for used cars
 - simple model
 - market breakdown

- Problems of **adverse selection** stem from asymmetric information that is present already before contracts are offered and accepted.
- The problem arises because informed parties' information about their "quality" affects their market behaviour and uninformed parties take this response into account.
- We will see that the market equilibria often fail to be Pareto optimal/efficient.
- Markets may shut down partially or even fully, despite potential gains from trade

Example: Akerlof's market for lemons

- Used cars can be either of good quality or a “lemon”
Seller knows quality, buyer does not (hidden information)
- At a given price, a seller will only supply if the quality of his car is sufficiently low (valuation is connected to quality)
⇒ bad quality drives out good quality
- Trade only takes place if buyers value cars much more than sellers. There is also the possibility of a complete market breakdown

Akerlof's market for lemons

- Consider a market for used cars. There are 150 risk-neutral sellers who want to sell a car each.
- 50 of these cars are in good condition. 100 of these cars are broken and need to be fixed.
- There are 160 potential risk-neutral buyers who want to buy at most one car each.

The valuations (willingness to pay/sell) are as follows: Outside option for potential buyers: $\underline{u} = 0$.

Condition of the car	Value for buyer	Value for seller	Share
Good	EUR 24.000	EUR 18.000	1/3
Bad	EUR 12.000	EUR 6.000	2/3
Expected value			

Complete and symmetric information

- Consider the market with complete and symmetric information.
- What does efficiency require?
- Note that each car is worth more to a buyer than to a seller.
- Hence, the market allocation is efficient if and only if all cars are sold.
- Under complete information (both buyers and sellers observe the quality of a car) this will be the case, and the prices are: (who has the market power?)

Incomplete but symmetric information (ISI)

- Now consider the market with incomplete, but symmetric information (ISI): neither sellers, nor buyers know the quality of a car (only the distribution over good and bad cars). Is the market outcome efficient?
- The expected value of a car for a seller is:
- If the market price weakly exceeds this number, the seller is ready to sell his car.
- The expected value of a car for a buyer is:
- If the market price is weakly lower than this number, the buyer is ready to buy the car.
- Since sellers have the market power, all cars will be sold at price

Efficiency (ISI)

Efficiency?

- As in the first best all cars have been sold.
- Some buyers paid 16.000 even though they end up with a car they value at 12.000 (if they knew the quality)
- Some sellers sold at price of 16.000 even though they value the car at 18.000 (if they knew the quality)
- Nevertheless, the allocation is efficient: all cars should be sold and they are.
- Prices solely affect income distribution, not efficiency (hence we don't worry about ex-post regret we simply analyse whether trade reaches first best volume).

Asymmetric information - Supply

- Suppose now only sellers know the quality of the car.
- We determine supply and demand functions and look for equilibrium
- Supply function:

$$S(p) = \begin{cases} 0 & \text{if } p < 6.000 \\ 100 & \text{if } 6.000 \leq p < 18.000 \\ 150 & \text{if } p \geq 18.000 \end{cases}$$

Asymmetric information - Demand

- Buyers can update their expected valuation based on the price: they know that only bad cars are sold at low prices
- If the price falls below 18.000 only bad cars are offered; buyers value them at 12.000. Hence for prices above 12.000 and below 18.000 no cars are demanded
- If the price exceeds 18.000 good and bad cars are offered. Buyers expected valuation is then 16.000, which is less than 18.000. So again no cars are demanded.
- Therefore, the demand function is

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$$D(p) = \begin{cases} 160 & \text{if } p \leq 12.000 \\ 0 & \text{if } p > 12.000 \end{cases}$$

Equilibrium

Equilibrium price is 12.000 and only the bad cars are sold
Inefficient!

- The good cars are not sold even though buyers value them more than sellers.
- **Good cars are driven out of the market. The lower the price, the lower is the average quality of cars that are offered.**

- This effect is called **adverse selection**: bad quality forces good quality out of the market.
- The degree of adverse selection depends on
 - ① the probability distribution over quality and
 - ② the buyer's preferences.
- The market for a good may completely break down as a result of adverse selection.

- If quality is a continuous variable, trade may break down due to asymmetric information. Consider the following example.
- Let there be a continuum of sellers of mass 1 and a continuum of buyers with mass $I > 1$.
- Quality q is distributed uniformly on the interval $[0,1]$, so the density function is $f(q) = 1$ on $[0,1]$.
- A buyer's valuation of a car of quality q is $\frac{3}{2}q$, so if a buyer gets a car of quality q at price p , her utility is $U^B = \frac{3}{2}q - p$, otherwise her utility is 0.
- If a seller trades her car of quality q at price p , her utility is p ; if she does not sell the car, her utility is q .
- Hence, sellers are willing to trade their cars if and only if $p \geq q$.

Asymmetric information

- Buyers take sellers behaviour into account. They are risk neutral and thus only care about expected quality.
- Determine the average quality at price p , i.e., $E(q|q \leq p)$

$$E(q|q \leq p) = \frac{\int_0^p [qf(q)]dq}{F(q = p)}$$

- Are consumers willing to trade at a positive price p ?
- What is the market equilibrium?

- What if $U^B = 2q - p$?

- Adverse selection may lead to an increasing demand function:

$$\frac{\partial D(p, E(q|q \leq p))}{\partial p} = \frac{\partial D}{\partial p} + \frac{\partial D}{\partial q} \frac{\partial E(q|q \leq p)}{\partial p}$$

- Note that $\frac{\partial D}{\partial p} < 0$, while $\frac{\partial D}{\partial q} > 0$ and $\frac{\partial E(q|q \leq p)}{\partial p} > 0$.
- There can be multiple equilibria with varying equilibrium prices.